Constraint Collapse: Why Systems Stop Correcting Themselves

Reality Drift Failure Modes Series | 03 Loss of Correction Binding

A. Jacobs — Reality Drift Framework

Core Claim

Systems do not fail because they make errors. They fail because they lose the ability to correct those errors. When constraint structures weaken, feedback stops binding behavior to reality.

The Pattern

At first, constraints are implicit. Norms, judgment, and friction shape behavior without needing to be formalized, acting as a stabilizing force that keeps metrics tied to reality. Metrics work as long as they sit inside this larger ecology, serving as rough indicators rather than definitions of value.

Under sustained competitive pressure, those constraints begin to be treated as inefficiencies. Friction is reduced and judgment is replaced with standardized metrics. Informal boundaries are removed in favor of scalable processes. As the system reorganizes, the metric shifts from measurement to objective.

Once that happens, the proxy slowly replaces the reality it was meant to represent. The system begins optimizing for the metric itself rather than the thing it was designed to capture. It can look increasingly successful by its own measures while becoming progressively disconnected from the underlying reality. What is lost is the structure that allowed the system to correct itself.

The Mechanism

Constraints are what make correction possible. They introduce cost, resistance, and consequence. They force systems to respond to reality rather than simply continue operating.

When constraints weaken, a shift occurs:

- Feedback becomes less binding
- Errors no longer interrupt behavior
- Metrics replace judgment
- Continuation becomes easier than correction

This creates a system that can absorb error without changing direction. Instead of being corrected, it adapts around the error and continues. Over time, correction stops being part of the system's operation and becomes external, optional, or impossible.

Where This Shows Up

In AI systems, outputs are optimized for coherence and usefulness, but without strong grounding constraints, errors don't reliably trigger correction. That same dynamic shows up in organizations, where processes continue even as outcomes degrade because internal metrics and incentives reward continuation. In media systems, engagement loops reinforce themselves without mechanisms for recalibration, and in individual cognition, high-information environments weaken the ability to filter, prioritize, and correct internal models.

Across domains, the same pattern appears. Systems continue operating even when they are no longer aligned.

Reality Drift Connection

Reality Drift describes what happens when systems remain operational while losing alignment with reality.

Constraint collapse is what allows that condition to persist. Without constraints, there is nothing to force the system back into alignment. Feedback no longer binds and correction no longer occurs.

The system can continue operating while drift becomes harder to reverse.

Core Framework and Sources

- Substack (Articles)
- GitHub (Full Library)
- DOI (Research Paper)
- Glossary & Definition